

Routing Algorithm Comparison

Algorithm	Finds Optimum?	Time Complexity	Space Complexity	Key Characteristic
DijkstraFlood	Yes	$O((V+E) \log V)$	$O(V)$	Full shortest-path tree
Dijkstra (mod.)	Yes	$O((V+E) \log V)$	$O(V)$	Halts at finish node
A*	Yes*	$O(E)$	$O(V)$	Heuristic reduces nodes visited
Hierarchical	Approx.	$O(\log V)$ eff.	$O(V)$ per level	Multi-graph, parallelisable
BiDirectional	Yes	$O((V+E)/2 \log V)$	$O(V)$	Two simultaneous searches